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A Model Predictive Control Algorithm for Autonomous Mining Dump Trucks

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Abstract: Mines have a significant impact on increasing the gross production and per capita income of a country. However, the mining environment is hazardous, and increasing productivity and efficiency while ensuring safety is crucial. One way to achieve this goal and reduce human fatalities is to use autonomous dump trucks. A suitable lateral controller for a dump truck can ensure that the truck moves on the desired path, increasing safety and productivity. This paper focuses on controlling mining dump trucks. A model predictive control based on the lateral model of a dump truck is designed to achieve the control objective. Some of the parameters of the dynamical model are estimated using the least squares method. Finally, an approximate model of an actual dump truck, the Sarcheshmeh mine environment, and the proposed controller are implemented in the Unreal Engine simulation environment to evaluate the designed controller.

Keywords: Autonomous Dump Truck, Model Predictive Control, Parameter Estimation, Unreal Engine.

1. INTRODUCTION

Mines are widely considered to be crucial assets for any country. Mining activities are a key determinant in enhancing industrial self-sufficiency, creating productive employment, and increasing the gross national product and per capita income of a country. Therefore, increasing productivity and improving the efficiency of hauling systems in mines is essential. The use of autonomous vehicles in mines is necessary due to adverse and hazardous working conditions, as well as the need to increase labor productivity (Dubinkin et al. (2020)). Moreover, eliminating human involvement from the production process enhances safety and efficiency, reducing operational costs (Voronov et al. (2020)).

Today, autonomous hauling systems in mines have gained significant importance. Autonomous hauling systems, also known as driverless systems, have already been utilized in surface mines for over a decade. In such mines, often referred to as smart mines, most equipment is automatically or remotely controlled by advanced control devices and algorithms (Voronov et al. (2020)). In recent years, major mining companies have taken steps to automate trucks that haul large loads and minerals. These actions eliminate the need for human operators, resulting in reduced labor costs, increased working hours, and improved transportation system efficiency (Ishimoto and Hamada (2020)).

On the other hand, extensive research has been conducted on autonomous vehicles in urban areas over the past decade. The concept of vehicle automation dates back to 1918 (Faisal et al. (2019)). The first autonomous car was invented by Norman Bel Geddes in 1939. In 1958, steering control of vehicles was processed by sensor information, which was known as sensor detecting. Over time and with technological advancements, autonomous vehicles have been able to gather more information about their surroundings using advanced sensors and react to them quickly (Faisal et al. (2019)).

One important aspect of autonomous vehicles is the modeling and control of vehicle motion along the intended path. There are several methods available for achieving this objective. In Lal et al. (2017), a geometric control method clled pure pursuit was introduced, which has demonstrated acceptable results. Another controller based on the geometric model is the Stanley controller, which usually is used for urban vehicle control due to its simplicity. The Stanley controller was investigated in Tippannavar et al. (2023) and tested using the CarSim simulation software. Furthermore, the performance of sliding mode and backstepping control techniques at high speeds were verified using the CarSim simulator (Norouzi et al. (2019)).

Other conventional control methods in autonomous vehicles such as PID and model predictive controller (MPC) have also been studied (Falcone et al. (2007); Borrelli et al. (2005)). According to the results obtained in Varma et al. (2020); Liu et al. (2021), the MPC method outperforms PID, LQR, Stanley, and pure pursuit controllers.

Some of these controllers have also been implemented for autonomous mining dump trucks. For instance, In

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one of the latest research studies, feedback control and modified Stanley methods were utilized for lateral control and path tracking of dump trucks (Chen et al. (2023)). Another approach used in dump trucks is optimal control, which has been evaluated using nonlinear H_∞ in Rigatos et al. (2020). Nonetheless, published results on the control methods used in mining dump trucks are very limited due to their intellectual property and commercial value.

Many modern mines, including the open-pit Sarcheshmeh copper mine in the Kerman province, Iran, require automation of their transportation systems and dump trucks. The control system of the dump truck's motion path is a fundamental component of automation. However, the existing controllers available for dump trucks face limitations in the steering angle and its changes and cannot change it arbitrarily. In this regard, a model predictive control (MPC) is proposed for path tracking of a dump truck, which has unique features such as the ability to impose comprehensive constraints on actuators, simultaneous decision-making with the extracted path, predicting future events, and taking appropriate control actions accordingly. In MPC, an optimization problem is solved at each stage within a limited time horizon to apply the optimal control input so that the predicted output of the system approaches the reference output as closely as possible (Varma et al. (2020); Chen et al. (2020)). MPC is a model-based approach, hence, a simple model of the intended dump truck is first extracted and linearized. Then, its parameters are determined and estimated. The use of MPC in this context can provide more efficient and effective control over the steering angle and improve the overall automation of the hauling system.

The proposed controller has not yet been implemented in a practical setting, although it is under work. Therefore, a reliable and realistic simulator is needed to assess the controller's performance. The Unreal Engine (UE) is a suitable choice due to its ability to simulate mining environments and implement dump trucks. Additionally, it can easily integrate with MATLAB software and the Simulink environment. Thus, the method developed in this paper is verified using UE simulation.

2. LATERAL DYNAMICAL MODELING OF A DUMP TRUCK

Different dynamical models yield different dynamic responses of a vehicle. It is important to choose the appropriate model for controlling the vehicle's motion. Since our objective here is the lateral control of the vehicle, a bicycle model is used to design the controller (Lin et al. (2019)).

The bicycle model for dump truck considers the movement of the vehicle in two dimensions, as depicted in Fig. 1: translational motion along the $X-Y$ axis and rotational motion around the Z axis. Since only the lateral model is of interest, other vehicle motions are not considered. Here, the position of the truck is denoted by X, Y in the global coordinates, and x, y in the local truck coordinates.

By applying Newton's laws and the equations obtained in Lin et al. (2019); Kebbati et al. (2023), the following differential equations are derived:

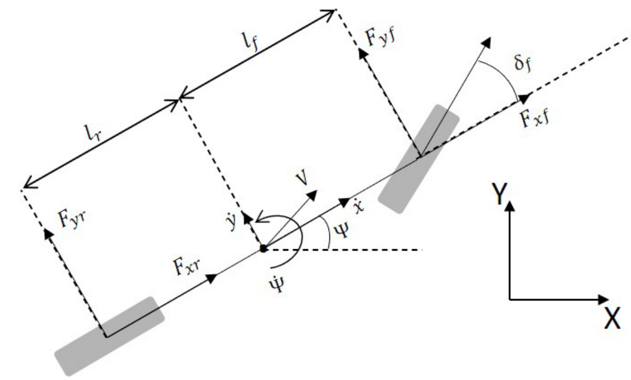

Fig. 1. The bicycle model for dump truck

$$
\begin{aligned}
\ddot{y} &= [\frac{(C_r l_r - C_f l_f)}{m\dot{x}} - \dot{x}]\dot{\psi} - [\frac{(C_f + C_r)}{m\dot{x}}]\dot{y} \\
&\quad + \frac{C_f}{m}\delta_f \\
\ddot{\psi} &= [\frac{-(C_f {l_f}^2 + C_r {l_r}^2)}{I_z \dot{x}}]\dot{\psi} + [\frac{(C_r l_r - C_f l_f)}{I_z \dot{x}}]\dot{y} \\
&\quad + \frac{C_f l_f}{I_z}\delta_f \\
\dot{Y} &= \dot{x}\sin\psi + \dot{y}\cos\psi \quad \xrightarrow{\text{for small } \psi} \quad \dot{Y} = \dot{x}\psi + \dot{y}
\end{aligned}
\tag{1}
$$

where $\dot{x}$ is the longitudinal linear velocity, $\dot{y}$ is the lateral linear velocity, ψ is the heading angle, $\dot{\psi}$ is the yaw rate, and δ_f is the steering angle. Also, m is the mass of the dump truck, l_f and l_r are the distances of the front and rear axles from the truck's center of mass, C_f and C_r are the cornering stiffness coefficients of the front and rear tires, and I_z is the moment of inertia about the z-axis.

The state vector is selected as $\mathbf{x} = [\dot{y} \;\; \psi \;\; \dot{\psi} \;\; Y]^T$, the input as $u = [\delta_f]$, and the output vector as $\mathbf{y} = [\psi \;\; Y]^T$. Notice that if these output signals, i.e., ψ, Y, are controlled, then x is successively is controlled as well. Also, for simplicity it is assumed that the longitudinal linear velocity $\dot{x}$ is constant. This is the assumptions often made in autonomous vehicles, as it allows utilization of single-variable control techniques for lateral and longitudinal control. In practice, often a PID controller is implemented for longitudinal control (i.e., a cruise control), following by a more advanced controller for lateral control.

The state equations, then, are

$$
\begin{aligned}
\dot{\mathbf{x}} &= A\mathbf{x} + Bu \\
\mathbf{y} &= C\mathbf{x}
\end{aligned}
\tag{2}
$$

where A, B, and C are given by:

$$
A = \begin{bmatrix}
\frac{-(C_f + C_r)}{m\dot{x}} & 0 & \frac{(C_r l_r - C_f l_f)}{m\dot{x}} - \dot{x} & 0 \\
0 & 0 & 1 & 0 \\
\frac{(C_r l_r - C_f l_f)}{I_z \dot{x}} & 0 & \frac{-(C_f {l_f}^2 + C_r {l_r}^2)}{I_z \dot{x}} & 0 \\
1 & \dot{x} & 0 & 0
\end{bmatrix}
$$

Table 1. Model parameters for BELAZ-7513 (available from manufacturer)

Description	Notation	Value/Unit
dump truck weight	m	110,000 Kg
front and rear axles distance	L	5.3 m
weight on the front axle	m_f	51,000 Kg
weight on the rear axle	m_r	49,000 Kg

$$B = \begin{bmatrix} \frac{C_f}{m} \\ 0 \\ \frac{C_f l_f}{I_z} \\ 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For a specific dump truck, the model parameters l_f, l_r, I_z, and cornering stiffness coefficients C_f and C_r are not directly available and need to be estimated. The following approximations can be used to estimate l_f, l_r, and I_z (Snider et al. (2009)):

$$\begin{aligned} l_f &= L\left(1 - \frac{m_f}{m}\right) \\ l_r &= L\left(1 - \frac{m_r}{m}\right) \\ I_z &= l_f l_r (m_f + m_r) \end{aligned} \tag{3}$$

Here, L is the distance between front and rear axles of the vehicle, m_f is the weight on the front axle, m_r is the weight on the rear axle, and m is the weight of the vehicle. These parameters are available for a specific dump truck, which in this research is a BELAZ-7513 mining dump truck with payload capacity of 130-136 tonnes. The parameters are reported in Table. 1 for BELAZ-7513. Then l_f, l_r, and I_z can be estimated using (3). The estimated values are reported in Table. 2.

For estimating the cornering stiffness coefficients, the least squares method is utilized. To start, the first two equation in (1) are rewritten as:

$$\ddot{y} + \dot{\psi}\dot{x} = \frac{1}{m}\left[(\delta_f - \frac{\dot{y} + l_f\dot{\psi}}{\dot{x}})C_f + \left(-\frac{\dot{y} - l_r\dot{\psi}}{\dot{x}}\right)C_r\right] \tag{4}$$

$$\ddot{\psi} = \frac{1}{I_z}\left[l_f(\delta_f - \frac{\dot{y} + l_f\dot{\psi}}{\dot{x}})C_f + l_r\left(\frac{\dot{y} - l_r\dot{\psi}}{\dot{x}}\right)C_r\right] \tag{5}$$

Since it may not be possible to directly measure $\ddot{y}$ and $\ddot{\psi}$, and only $\dot{y}$ and $\dot{\psi}$ may be measurable, (4) and (5) are discretized as:

$$\begin{aligned} &\dot{y}(k+1) - \dot{y}(k) + \dot{x}(k)\dot{\psi}(k)h \\ &\quad = \frac{1}{m}\left((\delta_f(k) - \frac{\dot{y}(k) + l_f\dot{\psi}(k)}{\dot{x}(k)})hC_f\right) \\ &\quad + \frac{1}{m}\left((-\frac{\dot{y}(k) - l_r\dot{\psi}(k)}{\dot{x}(k)})hC_r\right) \end{aligned} \tag{6}$$

$$\begin{aligned} &\dot{\psi}(k+1) - \dot{\psi}(k) \\ &\quad = \frac{1}{I_z}\left(l_f(\delta_f(k) - \frac{\dot{y}(k) + l_f\dot{\psi}(k)}{\dot{x}(k)})hC_f\right) \\ &\quad + \frac{1}{I_z}\left(l_r(\frac{\dot{y}(k) - l_r\dot{\psi}(k)}{\dot{x}(k)})hC_r\right) \end{aligned} \tag{7}$$

Table 2. Estimated model parameters

Description	Notat.	Value/Unit
C.G. distance to front axle	l_f	2.8 m
C.G. distance to rear axle	l_r	2.5 m
cornering stiffness of the front axle	C_f	943800 N/rad
cornering stiffness of the rear axle	C_r	1423200 N/rad

where h is the sampling time and k is the sampling index. With these equations in mind, the formulation for the least squares method is as follows:

$$\hat{A} = \hat{B}\begin{bmatrix} C_f \\ C_r \end{bmatrix} \tag{8}$$

where

$$\hat{A} = \begin{bmatrix} \dot{y}(2) - \dot{y}(1) + \dot{x}(1)\dot{\psi}(1)h \\ \dot{\psi}(2) - \dot{\psi}(1) \\ \vdots \\ \dot{y}(n) - \dot{y}(n-1) + \dot{x}(n-1)\dot{\psi}(n-1)h \\ \dot{\psi}(n) - \dot{\psi}(n-1) \end{bmatrix}$$

and

$$\hat{B} = \begin{bmatrix} \hat{B}_1 & \hat{B}_2 \end{bmatrix}$$

where

$$\hat{B}_1 = \begin{bmatrix} \frac{1}{m}\left[(\delta_f(1) - \frac{\dot{y}(1) + l_f\dot{\psi}(1)}{\dot{x}(1)})h\right] \\ \frac{1}{I_z}\left[l_f(\delta_f(1) - \frac{\dot{y}(1) + l_f\dot{\psi}(1)}{\dot{x}(1)})h\right] \\ \vdots \\ \frac{1}{m}\left[(\delta_f(n-1) - \frac{\dot{y}(n-1) + l_f\dot{\psi}(n-1)}{\dot{x}(n-1)})h\right] \\ \frac{1}{I_z}\left[l_f(\delta_f(n-1) - \frac{\dot{y}(n-1) + l_f\dot{\psi}(n-1)}{\dot{x}(n-1)})h\right] \end{bmatrix}$$

$$\hat{B}_2 = \begin{bmatrix} \frac{1}{m}\left[(-\frac{\dot{y}(1) - l_r\dot{\psi}(1)}{\dot{x}(1)})h\right] \\ \frac{1}{I_z}\left[l_r\left(\frac{\dot{y}(1) - l_r\dot{\psi}(1)}{\dot{x}(1)}\right)h\right] \\ \vdots \\ \frac{1}{m}\left[(-\frac{\dot{y}(n-1) - l_r\dot{\psi}(n-1)}{\dot{x}(n-1)})h\right] \\ \frac{1}{I_z}\left[l_r\left(\frac{\dot{y}(n-1) - l_r\dot{\psi}(n-1)}{\dot{x}(n-1)}\right)h\right] \end{bmatrix}$$

In the estimation of the cornering stiffness coefficients, 3500 data points for $\dot{y}$ and $\dot{\psi}$ are utilized. The estimated parameters are reported in Table 2.

3. DESIGN OF MODEL PREDICTIVE CONTROL

To design an MPC algorithm, time horizons and system samples are employed. Therefore, it is necessary to discretize the system model. Considering the system model in (2) and using Euler's method, the discretized model is:

$$\mathbf{y}(k+1) = A_d\mathbf{y}(k) + B_d u(k) \tag{9}$$

with

$$A_d = (I + hA). \quad B_d = hB, \quad C_d = C \tag{10}$$

here I represents the identity matrix.

The conventional state estimation equations obtained in Maciejowski (2002) for predicting future states with prediction horizon P. To simplify the calculations, the control horizon N_m is considered, indicating that the computed input only changes up to the control horizon and remains constant thereafter. i.e.,

$$u(k+i \mid k) = u(k+N_m-1 \mid k), \qquad N_m \le i \le P-1$$

or

$$\Delta u(k+i \mid k) = 0, \qquad N_m \le i \le P-1$$

Now considering the control horizon and after some manipulations we obtain:

$$u(k \mid k) = \Delta u(k \mid k) + u(k-1) \tag{11}$$

$$\begin{aligned}
\begin{bmatrix} \mathbf{x}(k+1 \mid k) \\ \vdots \\ \mathbf{x}(k+N_m \mid k) \\ \vdots \\ \mathbf{x}(k+P \mid k) \end{bmatrix} &= \begin{bmatrix} A_d \\ \vdots \\ A_d^{N_m} \\ \vdots \\ A_d^P \end{bmatrix} \mathbf{x}(k \mid k) \\
&+ \begin{bmatrix} B_d \\ \vdots \\ \sum_{i=0}^{N_m-1} (A_d^i B_d) \\ \vdots \\ \sum_{i=0}^{P-1} (A_d^i B_d) \end{bmatrix} u(k-1) \\
&+ \begin{bmatrix} B_d & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=0}^{N_m-1} (A_d^i B_d) & \sum_{i=0}^{N_m-2} (A_d^i B_d) & \dots & B_d \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=0}^{P-1} (A_d^i B_d) & \sum_{i=0}^{P-2} (A_d^i B_d) & \dots & \sum_{i=0}^{P-N_m} (A_d^i B_d) \end{bmatrix} \\
&\times \begin{bmatrix} \Delta u(k \mid k) \\ \vdots \\ \Delta u(k+N_m-1 \mid k) \end{bmatrix}
\end{aligned} \tag{12}$$

$$\begin{bmatrix} \mathbf{y}(k+1 \mid k) \\ \vdots \\ \mathbf{y}(k+P \mid k) \end{bmatrix} = \begin{bmatrix} C & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & C \end{bmatrix} \begin{bmatrix} \mathbf{x}(k+1 \mid k) \\ \vdots \\ \mathbf{x}(k+P \mid k) \end{bmatrix} \tag{13}$$

Combining (12) and (13), and using new notations we obtain:

$$\mathbf{y} = G_+ \Delta u_+ + \mathbf{f}. \tag{14}$$

The control law is then, as in Maciejowski (2002), obtained by optimizing the cost function:

$$J(\Delta u_+) = (\mathbf{y_d} - \mathbf{y})^T Q (\mathbf{y_d} - \mathbf{y}) + \Delta u_+^T R \Delta u_+ \tag{15}$$

where Q is the output weighting matrix, R is the input weighting matrix, and $\mathbf{y_d}$ is the reference trajectory. Calculating

$$\frac{\partial J(\Delta u_+)}{\partial \Delta u_+} = 0$$

yields

$$\Delta u_+ = (G_+^T Q G_+ + R)^{-1} G_+^T Q (\mathbf{y_d} - \mathbf{f}) \tag{16}$$

At each time step, the previous control command is added to Δu_+ and applied as the control command.

If there are constraints on the control commands, the optimization problem in (15) will no longer have an analytical solution and must be solved numerically using optimization algorithms. To impose constraints, the cost function is transformed into a standard quadratic equation as (Maciejowski (2002)):

$$J(\Delta u_+) = \frac{1}{2} \Delta u_+^T H \Delta u_+ + w^T \Delta u_+ + c \tag{17}$$

where

$$\begin{aligned}
H &= 2[G_+^T Q G_+ + R] \\
w &= -2[G_+^T Q (\mathbf{y_d} - \mathbf{f})] \\
c &= [(\mathbf{y_d} - f)^T Q (\mathbf{y_d} - \mathbf{f})]
\end{aligned} \tag{18}$$

For BELAZ-7513, based on the actual limitations of the steering system, the following constraints are considered:

$$\begin{aligned}
&-42 \le u_+ \le 42 \\
&-15 \le \Delta u_+ \le 15
\end{aligned} \tag{19}$$

Finally, the cost function in (17) can be numerically optimized subject to the constraints in (19). The objective is to find the optimal change in the steering wheel angle δ_f for following the desired path.

4. SIMULATION

4.1 Simulation Environment

To conduct a thorough simulation, a precise and authentic model of sensors (such as radar), a road model, and an autonomous vehicle model consisting of its dynamics and motion behavior is necessary (Schöner (2018)). To ensure the safe implementation of a truck in the real world, examining its movement in a simulated environment is important. A simulator provides a safe environment where the behavior of a truck can be investigated, which is similar to the real physical world. Game engines are some of the suitable tools for developing and building a simulator close to reality. One of the most well-known and advanced game engines is the Unreal Engine (UE) developed by Epic Games, which is used in this research.

For simulation, first, a 3D model of BELAZ-7513 dump truck was created, using the physical, geometrical, and dynamical parameters of the actual truck. Notice that UE utilizes complicated and accurate vehicle models much more involved than the linear model in (2) which we used for controller design. The complete model of the 3D environment of the open-pit Sarcheshmeh copper mine in Kerman province, Iran was also prepared and used for simulation. Moreover, various logic and control algorithms were implemented. Fig. 2 and Fig. 3 show the working environment and truck model within Unreal Engine. A snapshot of the final simulation environment can be observed in Fig. 4.

4.2 Simulation Results

To simulate the system, both the Unreal Engine (for simulating the truck and mine environment) and MATLAB

Fig. 2. Initial simulation environment and component editing within UE

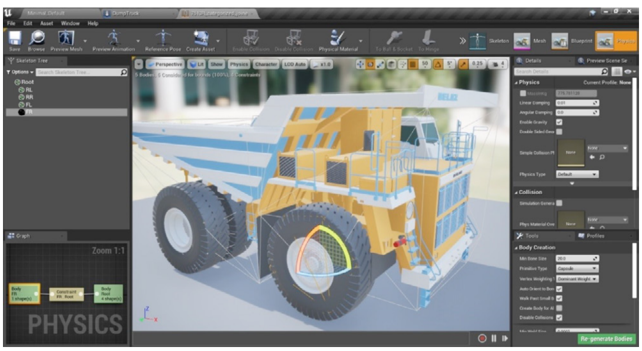

Fig. 3. The simulated truck body and its movable components such as tires

(for controller implementation) were utilized. The dump truck model in (2) whose parameters are given in Tables 1 and 2 was considered. The longitudinal speed of the dump truck is assumed to be constant (as it often is) and equal to 10 m/s. In simulation, a PID controller was designed and implemented to keep the longitudinal speed constant at 10 m/s. For lateral control, the MPC controller designed in this paper was implemented.

In each simulation step, the control signal was obtained by numerically optimizing the cost function in (17) subject to the constraints in (19). The control parameters are listed in Table 3.

In Fig. 5, the reference path (with relatively sharp turns for a dump truck) and the actual path of the dump truck are plotted. The lateral error is depicted in Fig. 6. Notice that the error peaks in the sharp turns.

The control command is also shown in Fig. 7. The negative and positive sign of the control command, represent clockwise and counter-clock-wise turning of the steering wheel. The longitudinal speed of the dump truck is demonstrates in Fig. 8. Simulation results show acceptable performance of the developed controller and satisfies the requirements of mine operations.

Table 3. Controller parameters

Name	Notation	Value/Unit
sampling time	h	0.016 s
output weighting matrix	Q	$0.8I$
input weighting matrix	R	I
prediction horizon	P	10
control horizon	N_m	3

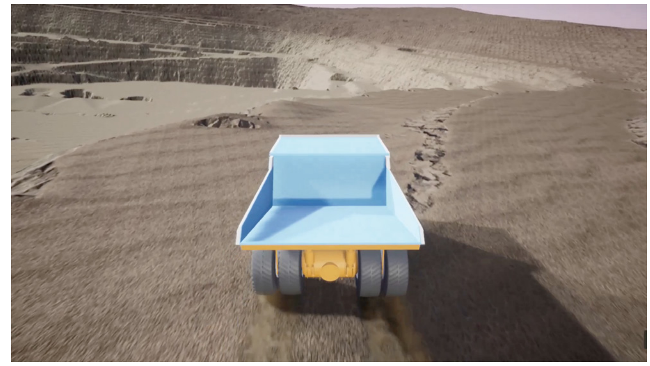

Fig. 4. Simulation environment within UE

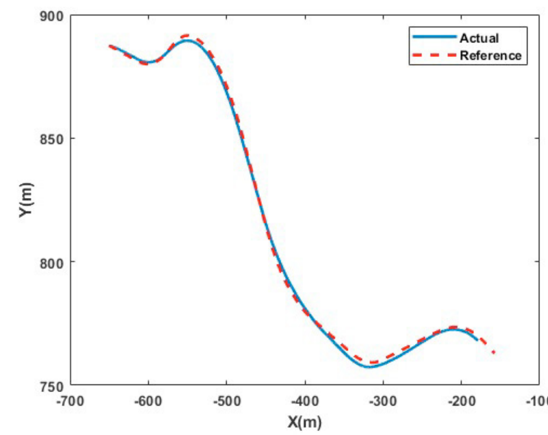

Fig. 5. Reference and actual paths of the dump truck

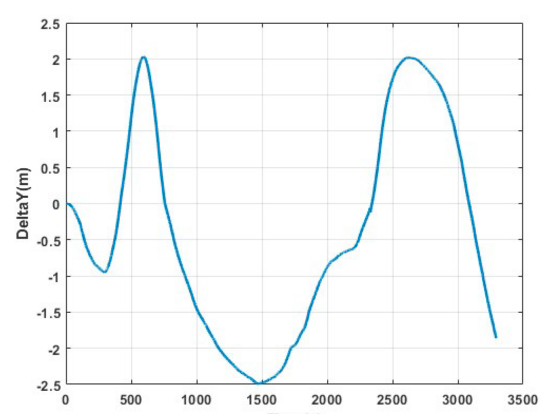

Fig. 6. Lateral error of the dump truck

5. CONCLUSION

The objective of this research was to design and implement an model predictive controller to control the lateral movement of mining dump trucks. To design a model predictive controller, the first step was to extract the dynamical model and estimate its unknown parameters. The least squares method was used to estimate some parameters that were not available from the manufacturer. The performance of the MPC was then evaluated using simulations conducted in the Unreal Engine and MATLAB environments, which showed satisfactory results in a realistic environment. The control signal was calculated subject to some

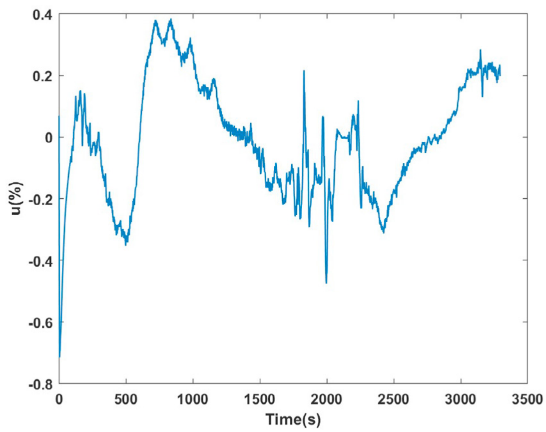

Fig. 7. Control signal

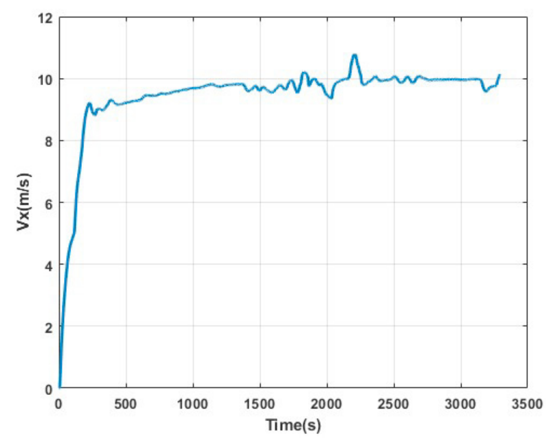

Fig. 8. Longitudinal speed of the dump truck

actual constraints imposed by the limitation of the steering mechanism of the truck. The proposed controller is under implementation on an actual autonomous dump truck, and the results will be reported in future publications.

REFERENCES

Borrelli, F., Falcone, P., Keviczky, T., Asgari, J., and Hrovat, D. (2005). MPC-based approach to active steering for autonomous vehicle systems. *International journal of vehicle autonomous systems*, 3(2-4), 265–291.

Chen, L., Qin, Z., Hu, M., Gao, H., Bian, Y., Xu, B., and Peng, X. (2023). Trajectory tracking control of autonomous heavy-duty mining dump trucks with uncertain dynamic characteristics. *Science China Information Sciences*, 66(10), 1–17.

Chen, S.P., Xiong, G.M., Chen, H.Y., and Negrut, D. (2020). MPC-based path tracking with pid speed control for high-speed autonomous vehicles considering time-optimal travel. *Journal of Central South University*, 27(12), 3702–3720.

Dubinkin, D., Sadovets, V., Syrkin, I., and Chicherin, I. (2020). Assessment of the need to create control system of unmanned dump truck. In *E3S Web of Conferences*, volume 177, 03022.

Faisal, A., Kamruzzaman, M., Yigitcanlar, T., and Curie, G. (2019). Understanding autonomous vehicles. *Journal of Transport and Land Use*, 12(1), 45–72.

Falcone, P., Borrelli, F., Asgari, J., Tseng, H.E., and Hrovat, D. (2007). Predictive active steering control for autonomous vehicle systems. *IEEE Transactions on control systems technology*, 15(3), 566–580.

Ishimoto, H. and Hamada, T. (2020). Safety concept and architecture for autonomous haulage system in mining. In *Proceedings of the International Symposium on Automation and Robotics in Construction (ISARC)*, volume 37, 377–384.

Kebbati, Y., Ait-Oufroukh, N., Ichalal, D., and Vigneron, V. (2023). Lateral control for autonomous wheeled vehicles: A technical review. *Asian Journal of Control*, 25(4), 2539–2563.

Lal, D.S., Vivek, A., and Selvaraj, G. (2017). Lateral control of an autonomous vehicle based on pure pursuit algorithm. In *IEEE International Conference on Technological Advancements in Power and Energy*, 1–8.

Lin, F., Chen, Y., Zhao, Y., and Wang, S. (2019). Path tracking of autonomous vehicle based on adaptive model predictive control. *International Journal of Advanced Robotic Systems*, 16(5).

Liu, J., Yang, Z., Huang, Z., Li, W., Dang, S., and Li, H. (2021). Simulation performance evaluation of pure pursuit, Stanley, LQR, MPC controller for autonomous vehicles. In *IEEE international conference on real-time computing and robotics (RCAR)*, 1444–1449.

Maciejowski, J. (2002). *Predictive Control: With Constraints*. Prentice Hall.

Norouzi, A., Masoumi, M., Barari, A., and Farrokhpour Sani, S. (2019). Lateral control of an autonomous vehicle using integrated backstepping and sliding mode controller. *Proceedings of the Institution of Mechanical Engineers, Part K: Journal of Multi-body Dynamics*, 233(1), 141–151.

Rigatos, G., Siano, P., Wira, P., Busawon, K., and Binns, R. (2020). A nonlinear h_∞ control approach for autonomous truck and trailer systems. *Unmanned Systems*, 8(01), 49–69.

Schöner, H.P. (2018). Simulation in development and testing of autonomous vehicles. In *Internationales Stuttgarter Symposium: Automobil und Motorentechnik*, 1083–1095.

Snider, J.M. et al. (2009). Automatic steering methods for autonomous automobile path tracking. *Robotics Institute, Pittsburgh, PA, Tech. Rep. CMU-RITR-09-08*.

Tippannavar, S.S., Yashwanth, S., Harshith, R., and Jain, S. (2023). Stanley controller based autonomous path planning and tracking in self-driving cars. *International Journal of Innovative Research in Advanced Engineering*, 10(03).

Varma, B., Swamy, N., and Mukherjee, S. (2020). Trajectory tracking of autonomous vehicles using different control techniques (PID vs LQR vs MPC). In *IEEE International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*, 84–89.

Voronov, Y., Voronov, A., and Makhambayev, D. (2020). Current state and development prospects of autonomous haulage at surface mines. In *E3S Web of Conferences*, volume 174, 01028.